



Blockchain–integrated IoT biometric architecture for secure and decentralized electronic voting

Tahzib Mahmud Rifat¹ · Md. Numanur Rahman¹ · Saurav Chandra Das¹ · Mahir Mahbub¹ · Md. Sanaullah² · Md. Ashiqussalehin¹ · Saad Aloteibi³ · Mohammad Ali Moni^{4,5}

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Abstract

A free and fair mass voting system is the key to all elections in the democratic world. Traditional voting systems, such as paper ballot voting, often face trust issues due to a lack of security and transparency. The electronic voting system (EVM) has emerged as an alternative with several advantages. However, EVM still presents its own set of challenges, including centralization, potential manipulation of software logic, accessibility issues, and a lack of verifiability, among others. To overcome these limitations, this study presents a blockchain-integrated IoT biometric voting architecture that coordinates low-cost biometric acquisition, multimodal voter verification, privacy-preserving pseudonymous identity handling, off-chain biometric protection, and smart-contract-based vote validation within a unified end-to-end election workflow. In the proposed architecture, the system integrates low-cost IoT devices for acquiring voters' biometric data (face and fingerprint), and robust voter verification is performed through AFIS-based fingerprint matching and InsightFace-based facial recognition with liveness-aware anti-spoofing checks. To strengthen privacy, the system avoids direct on-chain use of the voter's National ID and instead uses privacy-preserving pseudonymous identity management via encryption and hashing, while sensitive biometric records remain protected in off-chain storage. The proposed architecture defines three primary roles: admin, voter, and candidate. The entire process consists of two main phases: voter registration and voting. During registration, the administrator proceeds with voter registration using forms and biometrics. On election day, voter eligibility is confirmed by matching their live biometric input with the records created during registration. Each submitted vote is validated and written to the Ethereum blockchain via smart contracts. Experimental evaluation demonstrates that the system achieves high biometric performance, achieving 96.09% face-verification accuracy, a configured fingerprint FAR operating point of 0.0001 or 0.01%, an average IoT data transfer time of 6.4 seconds, which can be optimized to 3.2 seconds, a blockchain deployment cost of \$113.5, and a per-vote casting cost of approximately \$4.06 (at a baseline gas price of 20 Gwei). Combining IoT-based biometrics with blockchain enhances election security, protects privacy, and provides end-to-end transparency through a tamper-proof, trustworthy, and verifiable record of votes.

Keywords Blockchain-based e-voting · Privacy-preserving pseudonymous identity · Internet of things (IoT) · Multimodal biometric authentication · Anti-spoofing · Ethereum smart contracts

1 Introduction

Free and fair competitive elections are the foundation of representative democracy, as they enable citizens to freely choose their government and legitimize political authority through public consent (Gaspard 2025). A fair and transparent election ensures that people participate, are represented, and have the legitimacy of the government. In the context of a free and fair election, a secure, transparent, and accessible electoral process is the absolute minimum required to

uphold democratic principles and maintain public trust in electoral institutions. However, conventional voting systems, especially those based on ballot papers, face challenges due to a lack of confidence, voter impersonation, ballot manipulation, and other types of electoral fraud (Hossain Faruk et al. 2024). Additionally, they require substantial financial and time investments (Alam et al. 2018). Amid the pandemic crisis, many citizens lack enthusiasm to cast their votes on election day (Hanifatunnisa and Rahardjo 2017; Chen et al. 2020). To mitigate these issues, electronic voting (e-voting) has garnered significant attention worldwide as an alternative to conventional voting (Nikhare and Chandavarkar 2023).

Extended author information available on the last page of the article

Despite being a successful alternative, e-voting systems have vulnerabilities that may raise questions about the reliability and transparency of the voting process. The conventional technologies powering these systems, primarily centralized databases, are highly vulnerable to corruption and manipulation. In addition, cyber threats also raise significant risks that compromise the integrity of the voting process (Chaeikar et al. 2021). Addressing these weaknesses requires both decentralized record-keeping and strong, user-friendly voter authentication. The integration of blockchain-based dApps and biometrics offers promising solutions to address the limitations of existing e-voting systems. Blockchain technology promises reliability due to its decentralized, immutable, and transparent nature, enabling secure transaction recording without a central authority. By leveraging the unique features of blockchain, the ongoing challenges of existing voting systems can be addressed, resulting in a more transparent, secure, and trustworthy system (Berenjestanaki et al. 2024).

Over the past few years, several blockchain-based e-voting systems have emerged. Esposito and Choi (2023) proposed an Algorand-based e-voting framework that leverages the Pure Proof-of-Stake (PPoS) consensus to ensure scalability, fairness, and voter anonymity while minimizing the risk of forking and centralization. Somasekhar et al. (2024) proposed a blockchain-enabled digital voting system integrated with the InterPlanetary File System (IPFS) and the Practical Byzantine Fault Tolerance (pBFT) algorithm to enhance storage efficiency, reduce latency, and prevent vote manipulation. Indrason et al. (2025) introduced MBCSD-IoT. This multi-level Blockchain-assisted SDN-based IoT architecture integrates elliptic curve cryptography and hierarchical blockchain layers to secure large-scale voting processes across booth, district, and state levels. Razaque et al. (2025) presented an innovative contract-based e-voting system employing a Prioritized Delegated Proof-of-Stake (PDPoS) consensus to improve scalability and fairness among voters of different priority levels, along with Nightshade sharding to enhance throughput and reduce transaction delays. A different approach is presented in hybrid cloud-blockchain models proposed by Jayakumari et al. (2024), which aim to mitigate high costs but reintroduce centralization risks.

Recent work on blockchain-based e-voting increasingly integrates IoT endpoints with biometric authentication to enhance eligibility checks while maintaining ballot integrity. Alam et al. (2018) proposed a device-centric design (utilizing Raspberry Pi readers) with a permissioned blockchain, where a ballot UI unlocks only after local biometric/ID checks, and then commits an immutable, one-vote-per-voter record. Ibrahim et al. (2021) designed and developed a fingerprint-integrated prototype that performs batched voting using Merkle trees and disseminates blocks across

IoT stations for enhanced resilience. Rathee et al. (2021) demonstrated that coupling IoT edge collection with a dual-chain design (institutional records and cast status) improves traceability across election tiers and makes any interference evident throughout the system. Adeniyi et al. (2024) developed a blockchain-based e-voting system in which fingerprint minutiae features, extracted using Otsu thresholding and Zhang–Suen skeletonization, generate a 256-bit private key and an ECDSA-based public key, enabling biometric-driven authentication and immutable vote recording via smart contracts on a decentralized DApp. Prabakar and Kanchana (2025) proposed another IoT-focused work that explores multi-factor flows QR code factors with blockchain logging, custom ECC encryption, and high-throughput hashing to balance transparency with privacy via authority management.

Although several recent studies have explored blockchain-based e-voting systems with biometric integration, most fail to address all the vulnerabilities and procedural weaknesses present in traditional voting mechanisms. Some fail to integrate biometrics properly from end to end. Others come with overly complicated user interfaces or depend too much on specific hardware. And then there are the significant concerns, privacy risks, and scalability issues that hinder these systems from functioning smoothly in a real-time environment (Ohize et al. 2025). Comparative analyses (Berenjestanaki et al. 2024; Jafar et al. 2022) reveal significant gaps between theoretical models and prototypes. While privacy-preserving proofs demonstrate strong guarantees, few studies report real-world performance metrics. Empirical deployments, such as Ethereum-based pilots (Malkawi et al. 2023), highlight trade-offs: transparency is improved, but privacy leakage persists due to transaction traceability. Blockchain-based systems are vulnerable to Sybil, DoS, and 51% attacks (Salman et al. 2023). Therefore, it becomes clear that blockchain alone cannot guarantee a fully reliable e-voting framework. To achieve both security and usability, it must be complemented by strong voter authentication mechanisms that eliminate impersonation risks and enhance trust in the system.

Biometric identification, using fingerprints and facial recognition, provides a reliable way to verify voters' identities. By assigning a unique biometric signature to each voter, the e-voting system can effectively prevent impersonation and duplicate voting, and also ensure that only eligible voters can cast their votes. By integrating biometrics with blockchain, a robust voting framework is created that is not only more secure and reliable but also more inclusive and accessible. In addition, combining the machine learning approach with the biometric identification system increases the anomaly detection to a new level. It helps to identify real or fake fingerprints made of silicone or gelatin, as well as counterfeit faces, thereby improving the system's efficiency and preventing impersonation or fake voting

(Jafar et al. 2024). However, multi-modal biometric verification requires careful design to handle noisy or low-quality captures. A study shows that the open-source minutiae-based fingerprint identification system SourceAFIS performs well in resource-constrained deployments in controlled environments due to its small-size standard template generation, rather than working on raw fingerprint images. Fingerprint matching becomes faster than image-based matching techniques (Dhas et al. 2019). For facial verification, we have studied one of the best open-source models, InsightFace, a CNN-based embedding model that has demonstrated high accuracy even under challenging conditions, such as masked-face recognition (Wanyoni and Celik 2022; Deng et al. 2021; Das et al. 2026). However, most of these biometric models are tested in controlled environments rather than real-world electoral settings. As a result, issues such as scalability, liveness detection, and privacy-preserving data handling remain largely unaddressed in existing studies. Table 1 provides a detailed comparative analysis of the outcomes of various blockchain platform integrations into e-voting systems.

Although blockchain-based e-voting systems and biometric authentication techniques have been individually explored, several critical gaps remain in existing research:

- **Lack of robust end-to-end multimodal biometric verification:** Many blockchain-based e-voting systems improve decentralization and transparency, but they do not integrate multimodal biometric authentication (such as fingerprint and face recognition) throughout the complete voting workflow. In addition, most existing designs rely on a single biometric verification point, which increases vulnerability to faulty decisions, spoofing attempts, a single point of failure, and duplicate voting.
- **Limited practical deployment of low-cost IoT-enabled infrastructure:** A large number of proposed solutions depend on high-performance servers, specialized hardware, or overly complex architectures, which reduces their suitability for real-world election deployment. There is still insufficient attention to designing biometric verification workflows that can operate with affordable IoT-based acquisition devices while maintaining reliability, scalability, and accessibility.
- **Insufficient privacy-preserving identity handling and anti-spoofing protection:** Existing systems often do not adequately address two critical requirements at the same time: strong anti-spoofing protection and privacy-preserving identity management. Many works overlook liveness-based defenses against fake fingerprints and facial forgeries. In contrast, others continue to expose sensitive voter identifiers in blockchain interactions or do not clearly separate off-chain biometric storage from on-chain verification. This raises concerns regarding

biometric data leakage, identity exposure, and voter anonymity.

Despite the progress of blockchain-based e-voting research, there remains a practical technical need for a system that can simultaneously provide robust biometric voter verification, resistance to spoofing and duplicate voting, privacy-preserving identity handling, and transparent blockchain-based vote recording while remaining deployable with low-cost IoT-enabled infrastructure. Existing approaches typically satisfy only part of this requirement. Motivated by these limitations, our approach integrates multimodal biometric verification, consensus-based authentication, off-chain protection of sensitive biometric data, pseudonymous blockchain interaction, and smart-contract-based vote validation into a unified framework.

To address the identified practical and system-level engineering gaps, this paper makes the following contributions:

1. We present an end-to-end blockchain-integrated IoT biometric voting architecture that coordinates IoT-based biometric acquisition, multimodal biometric verification using AFIS and InsightFace, pseudonymous on-chain identity handling through encryption and hashing, and smart-contract-based vote verification within a unified election workflow. The proposed architecture is designed to support reliable voter authentication, eligibility verification, duplicate-vote prevention, and transparent result publication.
2. We develop a robust voter-authentication and vote-submission workflow that links live biometric verification with a derived pseudonymous voter hash for blockchain interaction. The workflow employs multiple parallel biometric verification servers and accepts a voter's identity only through a majority-consensus decision. Combined with liveness-aware anti-spoofing checks, this design improves authentication robustness, reduces dependence on a single biometric verification point, and enables the smart contract to verify eligibility and enforce one-person-one-vote control without storing raw voter identifiers, biometric templates, or sensitive identity records on-chain.
3. We implement the proposed architecture as a working engineering prototype and evaluate it across the biometric, IoT communication, and blockchain layers. The evaluation covers benchmark-based face verification performance, threshold-calibrated fingerprint authentication, IoT data transfer time, end-to-end voting latency, and smart contract gas cost analysis, thereby demonstrating the feasibility and practical deployment potential of the proposed system.

Table 1 Comparative Study of Blockchain-based E-Voting Systems

Ref.	Year	Technology Used	Contributions	Deployment	Issue Addressed	Key Findings
Salman et al. (2023)	2023	Blockchain (Ethereum focus)	Identifies common security attacks in blockchain e-voting	Simulation	Security threats (Sybil, DoS, manipulation)	No biometric integration, lacks defenses
Malkawi et al. (2023)	2023	Ethereum blockchain	Developed an e-voting system for Jordan's parliament elections	Prototype	Transparency, trust	No biometrics, gas fee issues
Esposito and Choi (2023)	2023	Algorand (PPoS)	Blockchain-based e-voting using Algorand	Prototype	Transparency, scalability	No biometric integration
Based et al. (2024)	2024	Hybrid blockchain, deep learning	Blockchain-based e-voting with hybrid learning	Prototype	Security, transparency	Limited biometric integration
Jayakumari et al. (2024)	2024	Hybrid Blockchain Cloud,	Proposed cloud-blockchain based e-voting	Prototype	Data integrity, transparency	No biometrics, cloud dependency
Somasekhar et al. (2024)	2024	Blockchain, PBFT, IPFS	Blockchain-based Byzantine tolerant voting	Prototype	Data integrity, tamper resistance	Complex consensus, no biometrics
Alzahab et al. (2024)	2024	Blockchain, Fuzzy commitment	Decentralized biometric authentication (fingerprints)	dApp prototype	Voter identity verification	Limited to fuzzy model
Adeniyi et al. (2024)	2024	Blockchain, PK cryptography	Fingerprint biometric-generated keys for e-voting	Prototype	Voter authentication	Limited to fingerprint
Nagaria and Shingadiya (2025)	2025	Blockchain, Smart contracts	Blockchain voting framework with evaluation metrics	Prototype	Transparency, secure tallying	Scalability issues
Razaque et al. (2025)	2025	Blockchain, DPoS	Prioritized smart contracts for scalable voting	Prototype	Scalability, secure contracts	No biometric integration
Prabakar and Kanchana (2025)	2025	Blockchain, IoT (QR codes)	IoT-based voting system using QR + blockchain	Prototype	Voter authentication, integrity	Weak biometric verification (QR only)
Peelam et al. (2025)	2025	Ethereum, Smart contracts	DemocracyGuard framework with face recognition	Prototype	Tamper resistance, transparency	Scalability limits, Ethereum dependency
Indrason et al. (2025)	2025	Multi-level Blockchain, SDN, ECC	Multi-level blockchain-assisted IoT voting	Prototype simulation	Hierarchical secure storage	Only simulation, complex infrastructure
Our Work		Ethereum, Smart contracts, AFIS, Insight-Face, IoT	End-to-end IoT-biometric-enabled secure blockchain-based e-voting system	Prototype (hardware, Anvil testnet)	Transparency, authentication, privacy preserving, real-time results	Prototype, no real-world deployment, depends on IoT/network stability, gas fees on public chain

2 System design and methodology

Traditional voting systems are time-consuming and require extensive voter verification. Before Election Day, authorities confirm voter addresses and identities using national identification cards. On the election day, voters must appear at their assigned polling stations to cast ballots. After voting, fingers are marked with ink to prevent duplication. Such centralized control often enables fraud or manipulation. Integrating blockchain plays a vital role in this context through its properties of immutability and decentralization. Once a vote is recorded, it cannot be altered. Also, smart contracts can help prevent duplicate voting. Moreover, decentralization ensures continuous operation even if some nodes fail, eliminating dependence on a single authority.

The proposed electronic voting system enhances transparency and fairness by combining IoT devices, biometric authentication, and blockchain technology. In traditional EVMs, where voters must go to specific polling booths, the proposed system allows voting from any authorized terminal nationwide, removing regional constraints. IoT devices capture and transmit biometric data, including fingerprints and facial information, to dedicated servers for verification. The architecture includes a blockchain server for the dApp and a distributed biometric-verification layer for authentication. Instead of relying on a single biometric server, the proposed system uses five parallel biometric verification servers for each modality (fingerprint and face). As illustrated in the Fig. 1, voters and administrators interact through a decentralized web-based application. Authentication is fully automated, ensuring that only eligible voters can participate after successful biometric verification.

2.1 Proposed system architecture

The proposed system operates in two phases: the pre-voting registration phase and the vote casting phase. The registration phase is conducted by the administrative authority to register voters and candidates. The authority collects voters' information through a dApp form and biometric devices. On the election day, voters go to the EVM booths to cast their votes. Before casting, they must verify themselves using a biometric identification system. When a vote is cast, it will be verified through smart contracts and stored in the Ethereum blockchain. The proposed e-voting system is designed to ensure secure, transparent, and verifiable elections by integrating blockchain and biometric technologies. The system incorporates four tightly integrated components: (1) blockchain-based vote storage and smart-contract validation, (2) multimodal biometric authentication using fingerprint and facial recognition, (3) privacy-preserving off-chain identity and biometric storage with pseudonymous

on-chain identity handling, and (4) a web-centric voting interface for administrative and voter interaction. The main stakeholders in our proposed system are voters and the election commission's administrative authority.

Here is a detailed description of the components introduced in the system:

- **Biometric-integrated registration:** The voters and candidates start their e-voting system journey by going through a registration process. The registration process collects general information, along with biometric data such as fingerprints and facial scans, and links it to the voter's National ID (NID) within the protected off-chain infrastructure. During this stage, the system also prepares a privacy-preserving pseudonymous identity for blockchain interaction, so that the voter's raw NID is not exposed directly on-chain. This process reduces the risk of duplicate registrations while maintaining data integrity and laying a trusted foundation for transparent elections.
- **Consensus-based biometric verification:** To improve authentication robustness, biometric verification is carried out by parallel verification servers that independently process the same biometric query. A voter's identity is accepted only when at least three of the five servers return the same NID, following a 3-out-of-5 majority-consensus rule. This design reduces reliance on any single verification point and strengthens resilience against faulty responses, spoofing attempts, and server compromise.
- **Off-chain server and blockchain storage:** Sensitive personal and biometric information, such as raw NID numbers, biometric photos, fingerprint templates, and encrypted salts, is stored and processed within the protected off-chain infrastructure. Keeping these records off-chain improves efficiency, reduces storage cost, and strengthens privacy protection. For blockchain interactions, the system derives a privacy-preserving, pseudonymous voter hash from the voter's identity record and uses that identifier for eligibility verification and vote status control. Meanwhile, the blockchain server handles smart-contract-based vote validation and immutable vote recording on the decentralized ledger. Together, these two layers improve privacy, security, and trustworthiness in the proposed voting process.
- **RESTful API Integration:** In the proposed e-voting system, the RESTful API integration helps as a communication bridge between the frontend dApp, the backend server, and the blockchain network. It performs registration, biometric authentication, and vote submission. For example, the frontend web application sends requests to the backend server for fingerprint authentication, and the backend server responds with a success code or an alert indicating failure. It ensures modularity, scalability, and

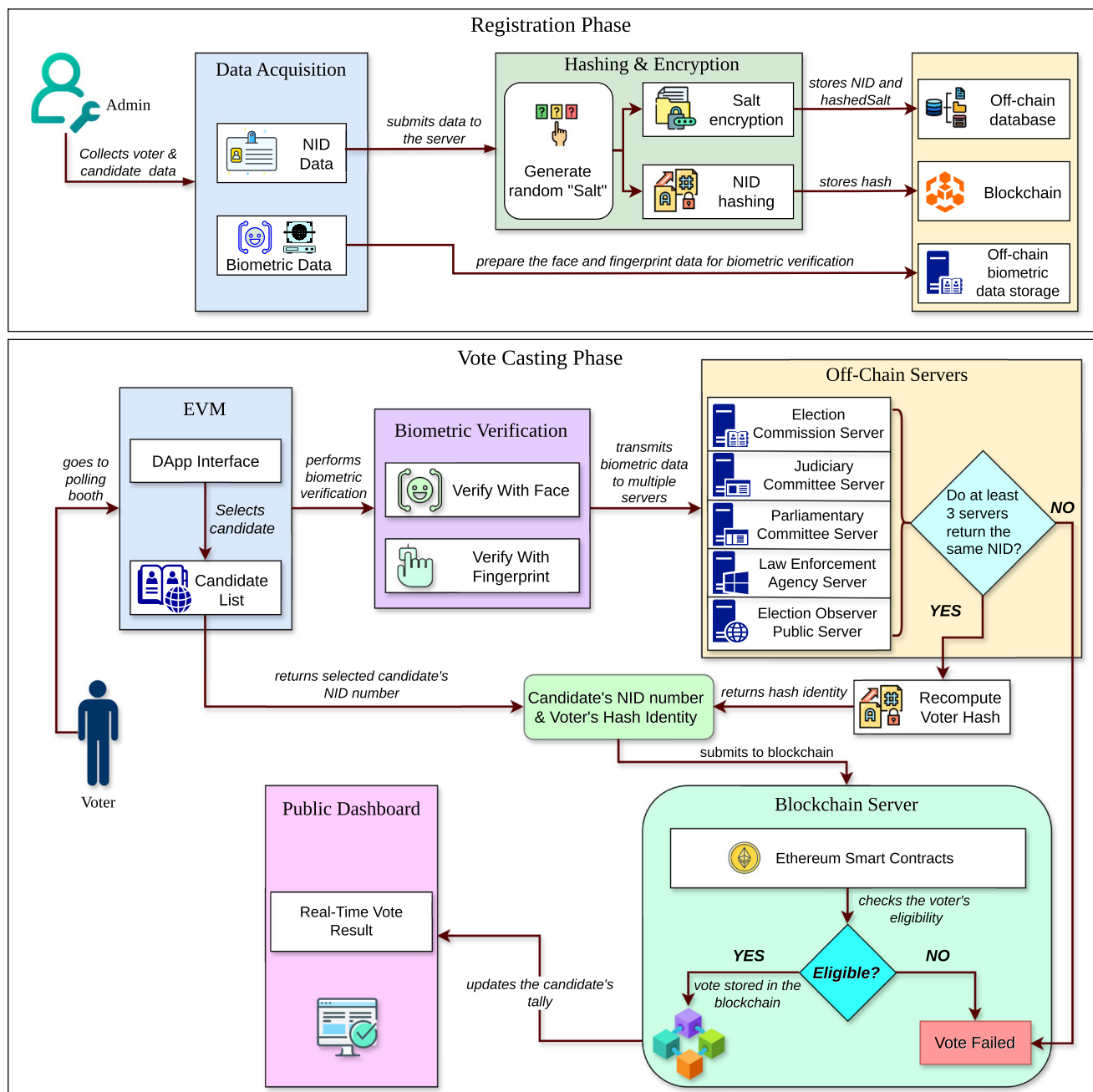


Fig. 1 Architecture of the proposed electronic voting system that highlights data acquisition, data encryption & hashing, biometric verification integration, blockchain-based voting, and real-time result publication

platform independence, which makes the system easier to maintain, update, and integrate with other components.

- **Admin dashboard and public dashboard:** The main stakeholders of the system are voters (public) and the administrative authority of the election commission. They both have their own dashboard. The admin dashboard allows the authority to manage the election process, including voter and candidate registration, deploying the election, and monitoring. The public dashboard ensures

transparency by displaying the real-time vote counts and final results. This helps verify election outcomes without accessing sensitive data.

To ensure the proper functionality of these components, the required tools and technologies work together across the different layers, maintaining system integrity. The seamless interaction among IoT devices, smart contracts, and biometric modules ensures a smooth, reliable voting experience.

Overall, these integrated components make the proposed system a robust, scalable, and privacy-preserving e-voting framework suitable for real-world deployment.

2.2 Workflow of the proposed system

The voting system is primarily divided into two phases: the pre-voting phase, also known as the registration phase, and the vote-casting phase. Each phase includes several activities and workflows to ensure the integrity and security of the entire voting process. The activities that take place in both phases can be described as follows:

2.2.1 Registration phase

The registration phase focuses on collecting voter and candidate data, enrolling them into the system, and configuring the platform in preparation for the election. Registration processes are completed well before voting day to ensure the entire system remains fully operational on voting day. Admin has major activities in this phase. The activities it includes:

- **Admin Authentication Process:** Admin starts his journey in this system by connecting his admin wallet. Only after connecting the wallet will he be able to access the admin dashboard. The admin dashboard is responsible for collecting and registering voter and candidate data. The DApp will then store that data on the corresponding servers.
- **Data Acquisition Process:** Voters and candidates have different sections for data collection and submission. Because candidates will provide other information to the voters. But in general, their data is collected using dApp forms. After filling out the form, the admin collects facial images and fingerprint data from voters using IoT devices connected to the dApp.
- **Registration Process:** After data acquisition, the admin submits the data to the dApp server. The DApp server then stores the data in the off-chain server and blockchain server.

Before Election Day, the authority prepares the servers for deployment. Figure 2 shows the workflow of the registration phase that captures the interaction between the system components involved in this phase.

Here, Algorithm 1 describes the voter registration process in the proposed system. The procedure includes admin wallet authorization, validation of voter details such as NID, name, and age, prevention of duplicate registrations, biometric enrollment using face and fingerprint data, generation of a random salt for each voter, secret-key-based protection of the salt, and derivation of a voter hash for blockchain registration. In this workflow, the voter's raw identity data and biometric

records remain in protected off-chain storage, while only the derived voter hash is recorded on the blockchain. In Algorithm 1, s_v denotes the unique random salt generated for voter v , $encSalt$ denotes the encrypted salt stored in the off-chain database, h_v denotes the derived voter hash recorded on the blockchain, and K_{sec} denotes the secret key used to protect the generated salt before storage.

Candidate registration is also completed during the pre-voting phase through the admin dashboard. In the current prototype, candidate records are verified by validating identity details, checking for duplicate entries in the off-chain candidate database, and registering the approved candidate identifier in the blockchain smart contract before election activation.

Algorithm 1 Voter Registration Algorithm.

```

1: function REGISTERVOTER(NID, Name, DateOfBirth, Location,
   FaceImages, FingerprintTemplate)
2:   if  $\neg \text{AdminWalletConnected}()$  then
3:     return "Unauthorized admin access"
4:   end if
5:   if  $\text{len}(\text{NID}) \neq 10$  or  $\neg \text{IsAlphabetic}(\text{Name})$  or
      $\text{Age}(\text{DateOfBirth}) < 18$  then
6:     return "Invalid voter details"
7:   end if
8:   if  $\text{NID} \in \text{voterCollection}$  then
9:     return "Duplicate NID found"
10:  end if
11:  faceStatus  $\leftarrow \text{CaptureFace}(\text{NID}, \text{FaceImages})$ 
12:  fingerStatus  $\leftarrow \text{EnrollFingerprint}(\text{NID}, \text{FingerprintTemplate})$ 
13:  if faceStatus  $\neq \text{success}$  or fingerStatus  $\neq \text{success}$  then
14:    return "Biometric capture failed"
15:  end if
16:   $s_v \leftarrow \text{GenerateRandomSalt}()$ 
17:   $encSalt \leftarrow \text{EncryptWithSecretKey}(s_v, K_{sec})$ 
18:   $h_v \leftarrow \text{Hash}(\text{NID} \parallel s_v)$ 
19:  if  $h_v \in \text{voterHashRegistry}$  then
20:    return "Duplicate voter hash found"
21:  end if
22:  Insert {NID, Name, DateOfBirth, Location, encSalt}
    into voterCollection
23:  Store FaceImages and FingerprintTemplate in
    offChainBiometricStorage
24:  Store  $h_v$  in voterHashRegistry
25:  return "Voter created successfully"
26: end function

```

2.2.2 Vote-casting phase

Once the registration phase is complete, the system is ready for election-day deployment. The authority deploys EVMs across the country. Admin has fewer activities on this day. The registered voters perform the main workflows. The admin logs in to the admin dashboard and makes the election live. The remaining workflows depend on the registered

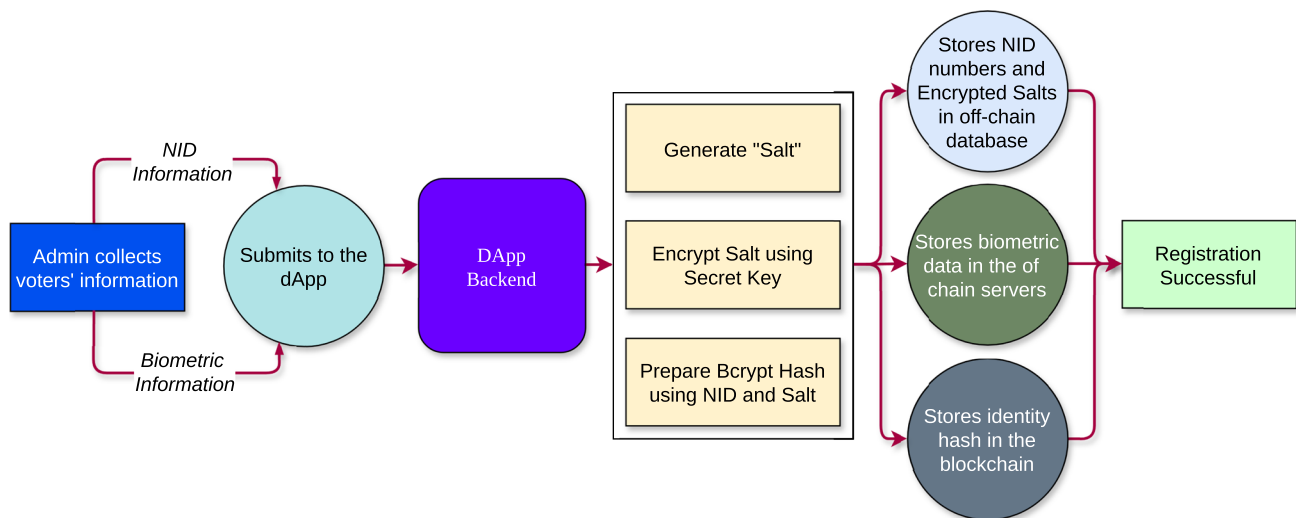


Fig. 2 The workflow diagram of the registration process. Voter information (NID and biometric data) is collected and submitted through the dApp. Biometric data is stored off-chain, while the hash of the voter's identity is recorded on the blockchain

voters. This phase is illustrated in the infographic in Fig. 3, which outlines the vote-casting process.

Vote-casting activities are as follows:

- **Biometric authentication & login:** Voters attempt login by providing either their face or fingerprint data for biometric verification. The submitted biometric query is processed through the authentication pipeline, and only

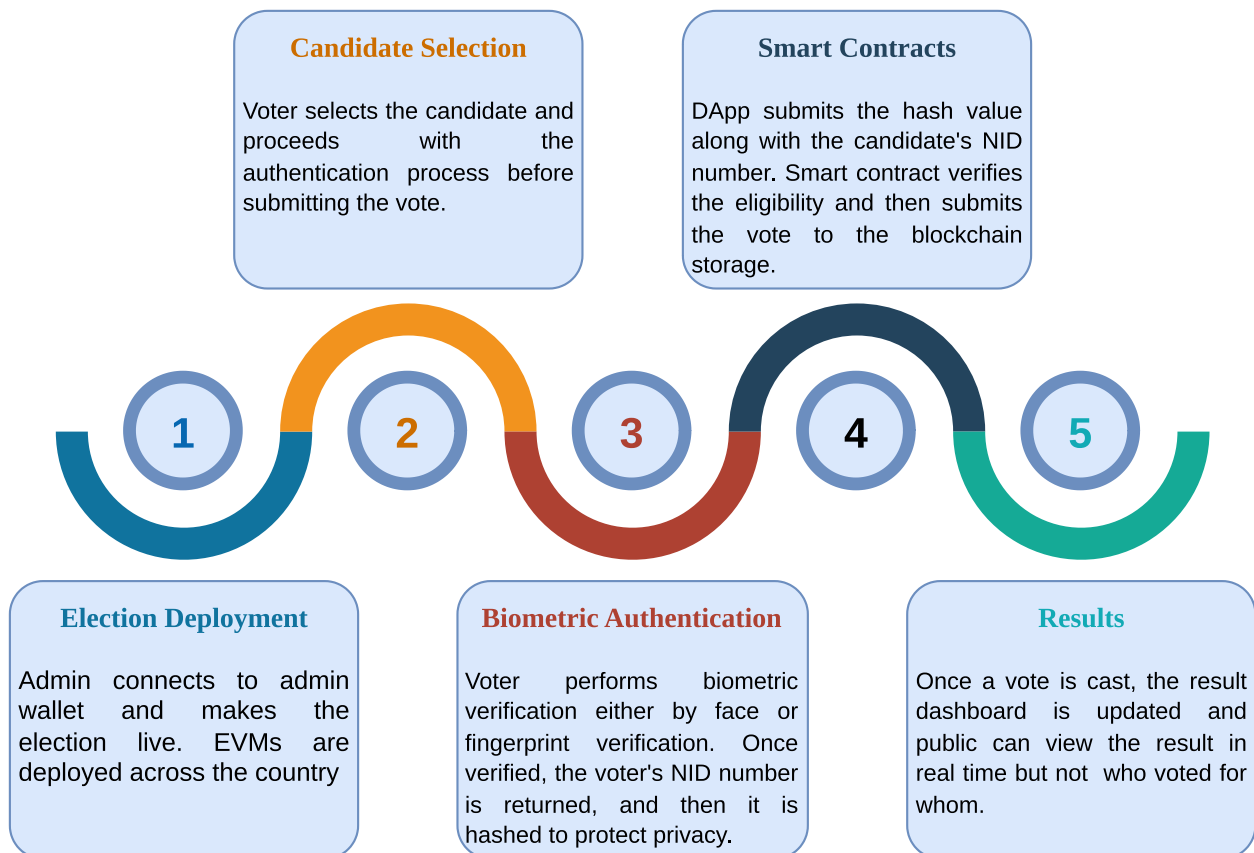


Fig. 3 Workflow diagram of the vote-casting process that highlights five main activities: election deployment, candidate selection, biometric authentication, smart contracts verification, and result viewing

after successful verification is the voter allowed to proceed. The backend then retrieves the voter's protected off-chain identity record, decrypts the stored salt, and recomputes the corresponding pseudonymous voter hash from the voter's identity information. This derived voter hash is then checked against the registered voter commitments stored on the blockchain. If the hash matches a valid, eligible record, the voter is allowed to log in and proceed to vote.

- **Vote-casting procedure:** After successfully logging in to the system, the voter dashboard is opened. The voter selects a candidate from the dashboard and submits the vote request. Instead of sending the voter's raw NID to the blockchain, the system submits the derived pseudonymous voter hash, along with the selected candidate identifier, to the smart contract. The vote is then sent to the blockchain for verification via the smart contract. The smart contract verifies that the submitted voter hash corresponds to a registered eligible voter, ensures that the same voter hash has not already been used, and checks that the selected candidate identifier is valid for the active election. If all conditions are satisfied, the contract records the vote on the blockchain, increments the selected candidate's vote count, and marks the voter's hash as used. Once a voter casts a vote, the corresponding voter-hash status is updated to indicate that the vote has already been cast. Then the voter is automatically logged out of the dApp to prevent integrity issues during voting.
- **Result viewing process:** There is a public dashboard to view the real-time election result. Everyone, including voters, observers, and third parties, can view the votes and tally. Blockchain transparency ensures tamper-proof and auditable election outcomes.

Here is algorithm 2 that describes the processes of vote casting activities on the election day.

2.3 IoT integrated biometric authentication procedures

Following the registration and voting processes, this section describes the biometric verification mechanism that integrates IoT devices with biometric algorithms in the proposed electronic voting architecture. To enhance security and reduce the risk of impersonation, a multimodal biometric authentication procedure is implemented that includes both fingerprint and facial recognition.

2.3.1 AFIS fingerprint identification process

The R307 optical fingerprint sensor, connected to the ESP32, serves as the point of fingerprint collection. The fingerprint

Algorithm 2 Vote-Casting Algorithm.

```

1: function VOTECASTING(CandidateID, BiometricQuery)
2:   Voter selects a candidate
3:   verifiedID  $\leftarrow$  BiometricAuthenticateWithConsensus(BiometricQuery)
4:   if verifiedID =  $\emptyset$  then
5:     return "Vote rejected"
6:   end if
7:   voterRecord  $\leftarrow$  GetProtectedOffChainRecord(verifiedID)
8:   sv  $\leftarrow$  DecryptWithSecretKey(voterRecord.encSalt, Ksec)
9:   hv  $\leftarrow$  Hash(voterRecord.NID || sv)
10:  if hv  $\notin$  voterHashRegistry then
11:    return "Vote rejected"
12:  end if
13:  Submit {CandidateID, hv} to castVote() in the smart contract
14:  if hv is already used or CandidateID is invalid then
15:    return "Vote rejected"
16:  else
17:    Record vote on blockchain
18:    Update selected candidate's voteCount
19:    Mark hv as used in blockchain state
20:    return "Vote confirmed"
21:  end if
22: end function

```

verification process is performed by the Automated Fingerprint Identification System (AFIS) server. The R307 sensor doesn't send a direct fingerprint image because of the risk of biometric data theft. Instead, it transmits feature data to the ESP32 microcontroller. The ESP32 transmits the data to an intermediary server for further processing before it reaches the AFIS server. We have developed an ExpressJS server that receives raw fingerprint data and converts it into a usable fingerprint image. This server then communicates with the AFIS server to perform the minutiae-based extraction. The AFIS server prepares a database containing templates for individual fingerprint images. We propose to use SourceAFIS, an open-source version of AFIS that performs minutiae-based template matching. When a fingerprint is captured, the server first enhances the image to improve ridge clarity and reduce noise. The fingerprints shown in Fig. 4 were captured by the R307 fingerprint sensor.

The AFIS server then extracts minutiae points and converts them into a compact ANSI/ISO 378 template of 1 KB, enabling efficient storage and deployment at scale. During enrollment, this template is stored securely in the AFIS database and linked to the voter's unique ID (NID). Here is the algorithm 3 that is followed in the fingerprint verification process.

On election day, when a voter performs fingerprint verification, the sensor captures the fingerprint and processes it to extract the template T_p from the image. The AFIS server initializes a matcher M . The matcher M sequentially compares the probe template T_p against all enrolled templates stored in the database DB to find the highest matching score *bestScore*. Before starting the matching process, the AFIS

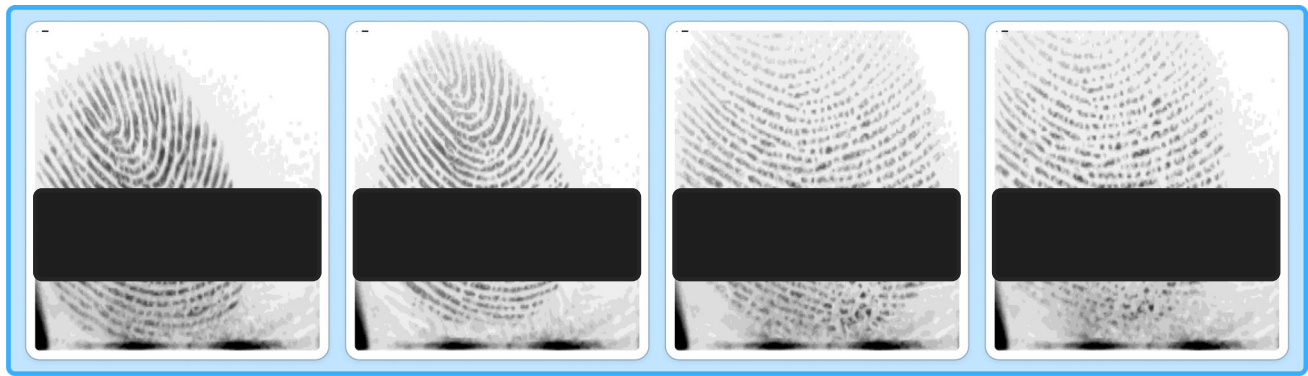


Fig. 4 Two fingers were enrolled for each voter, and each finger has two images for better accuracy. (Black rectangles are added to protect the privacy of the fingerprints)

already has a threshold score TH with a false acceptance rate (FAR) of 0.0001 or 0.01% for the enrolled templates in the database DB . The server then selects the individual with the highest similarity score, $voterID$, as the best candidate, $bestPerson$. Finally, a decision is made by comparing the highest score, $bestScore$, against a predefined threshold TH . If the best score, $bestScore$, exceeds the predefined threshold, TH , the algorithm accepts the match and returns the identifier $voterID$ along with the score $bestScore$. Otherwise, it rejects the probe as unidentified and returns null as the dApp's decision. The dApp then accepts or rejects the authentication attempt based on the decision. This ensures reliable fingerprint-based authentication in the proposed voting system.

These processes are visualized in the workflow diagram shown in Fig. 5, which outlines the integration of IoT devices into the fingerprint enrollment and authentication process.

Algorithm 3 Fingerprint Matching Algorithm.

```

1: function FINGERPRINTIDENTIFICATION( $I_p, DB, TH$ )
2:    $T_p \leftarrow \text{EXTRACTTEMPLATE}(I_p)$ 
3:    $M \leftarrow \text{MAKEMATCHER}(T_p)$ 
4:    $bestScore \leftarrow -\infty$ 
5:    $bestPerson \leftarrow \emptyset$ 
6:   for each  $voterID \in DB$  do
7:      $score \leftarrow \max\{\text{MATCH}(M, T) \mid T \in DB[voterID]\}$ 
8:     if  $score > bestScore$  then
9:        $bestScore \leftarrow score$ 
10:       $bestPerson \leftarrow voterID$ 
11:    end if
12:  end for
13:  if  $bestScore \geq TH$  then
14:    return (ACCEPT,  $bestPerson$ ,  $bestScore$ )
15:  else
16:    return (REJECT,  $\emptyset$ ,  $bestScore$ )
17:  end if
18: end function

```

2.3.2 InsightFace face recognition process

In our proposed multi-modal biometric authentication system, the proposed architecture also incorporates a CNN-based facial recognition system to ensure robust and accurate detection. We have introduced the InsightFace facial recognition system, which includes a face detector, an aligner, and an ArcFace recognizer. ArcFace is widely recognized for its discriminative power, which produces highly robust face embeddings that generalize well across variations in lighting, pose, and occlusion. To strengthen the robustness and efficiency of the face recognition system, we adopt a centroid-based similarity approach in the embedding space. During registration, each voter's facial image $\mathbf{x}$ is passed through the ArcFace embedding network $f_\theta(\cdot)$ to generate a 512-dimensional embedding vector:

$$\mathbf{e} = f_\theta(\mathbf{x}) \in \mathbb{R}^{512}. \quad (1)$$

For each enrolled voter y , we compute a class centroid μ_y as the mean of all their embeddings. At recognition time, the similarity score between a live embedding $\mathbf{e}$ and a stored centroid μ_y is measured by cosine similarity:

$$s_y(\mathbf{e}) = \frac{\mathbf{e}^\top \mu_y}{\|\mathbf{e}\|_2 \|\mu_y\|_2}. \quad (2)$$

The voter is then assigned the identity:

$$\hat{y} = \arg \max_y s_y(\mathbf{e}), \quad (3)$$

provided that the maximum similarity exceeds a threshold τ_{sim} .

The proposed centroid-based representation provides a compact and robust identity model for each voter. Unlike

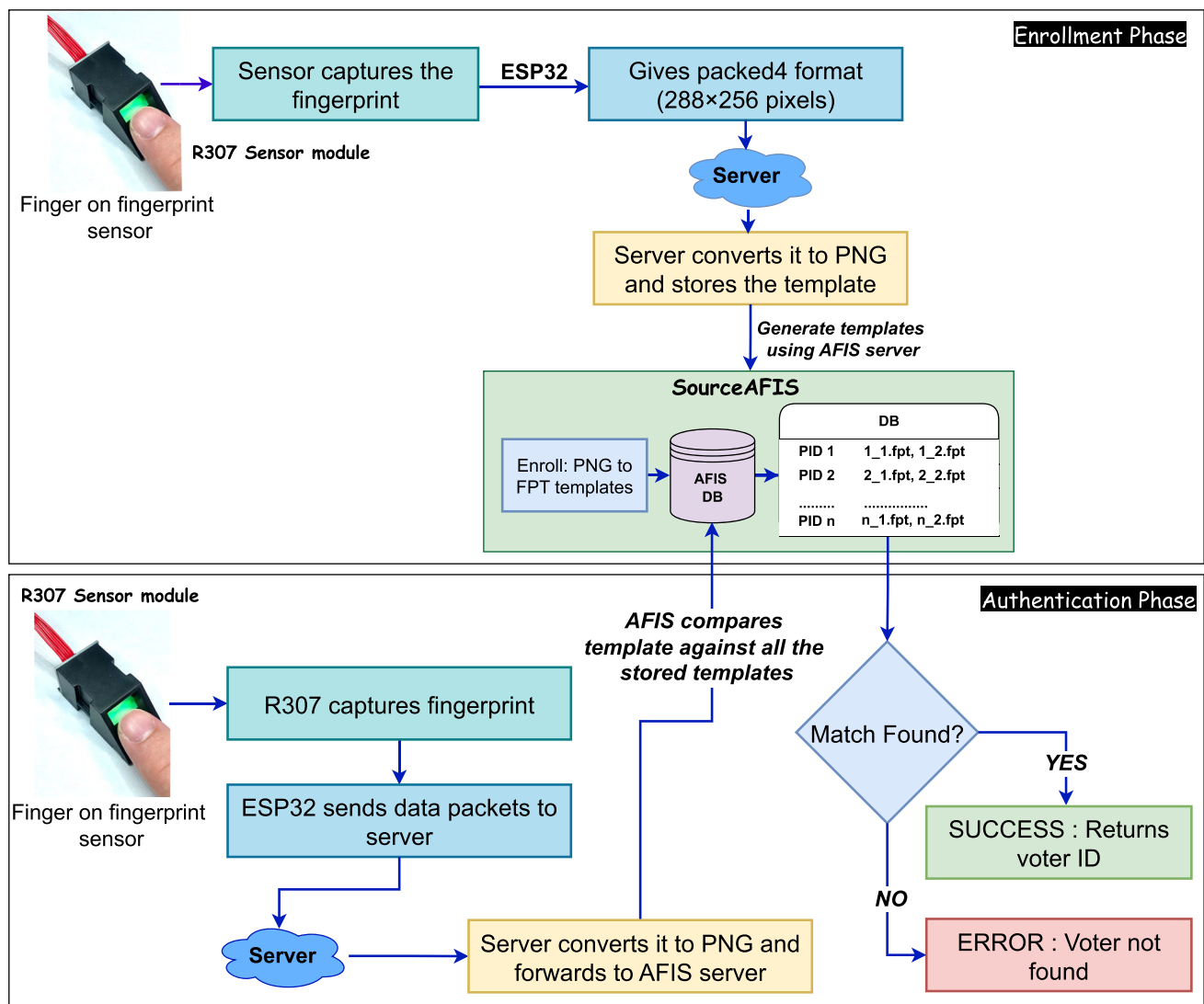


Fig. 5 Workflow diagram of the fingerprint authentication procedures that demonstrates the integration of IoT devices, data capture and transmission, AFIS integration, and verification process

model-based classifiers, this approach does not require training a global decision model and naturally supports incremental enrollment. When a new voter is registered, their centroid is calculated and added to the database without affecting existing identities. This eliminates the need for retraining and ensures scalability in large-scale election environments. Furthermore, cosine similarity is particularly well-suited for deep face embeddings, as such embeddings are typically normalized and optimized for similarity-based comparisons. This enables reliable matching even under variations in lighting, pose, and occlusion.

To ensure that recognition is not bypassed by spoofing or presentation attacks, the proposed system integrates liveness detection based on eye blinks and head movements. Eye blinks are measured using the Eye Aspect Ratio (EAR). A valid blink is confirmed when the EAR drops below a

closed-eye threshold and then rises above an open-eye threshold within a fixed duration. Only if the voter passes the required number of blinks and head-turn checks, and the face embedding is verified by cosine similarity exceeding a predefined threshold, is the voter's identity accepted.

Here is the notation Table 2 for the terms used in the equations:

The Fig. 6 represents the face recognition workflow in detail.

2.4 Blockchain workflow

In our proposed system, the blockchain operates like a digital ledger that cannot be altered or erased. Once something is written inside it, it stays there forever. This property is beneficial for elections because it ensures that votes cannot be secretly changed or deleted after being cast.

Table 2 Notation used in facial recognition and liveness detection equations

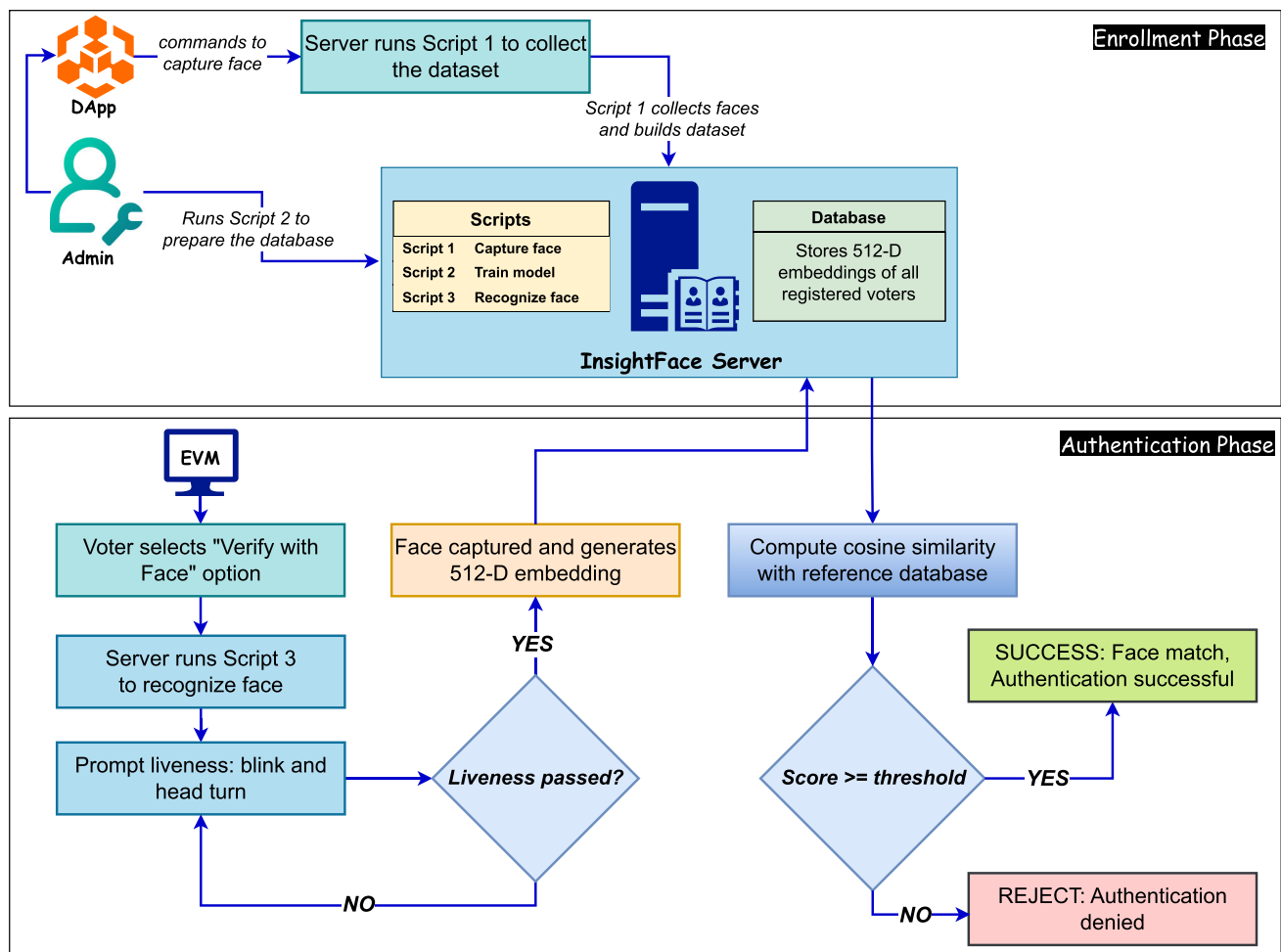
Symbol	Description
$\mathbf{x}$	Input facial image of a voter.
$f_{\theta}(\cdot)$	ArcFace embedding function parameterized by θ .
$\mathbf{e} \in \mathbb{R}^{512}$	512-dimensional face embedding vector produced from $\mathbf{x}$.
μ_y	Centroid (mean of embeddings) representing voter y in the feature space.
$s_y(\mathbf{e})$	Cosine similarity between embedding $\mathbf{e}$ and centroid μ_y .
$\ \cdot\ _2$	ℓ_2 -norm (Euclidean norm) of a vector.
$\hat{y}$	Predicted identity (class label) of the voter.
τ_{sim}	Similarity threshold for deciding whether a face is accepted.

Our system uses a single Solidity smart contract called *LocalVoting*. This contract was written, compiled, and deployed on the Ethereum blockchain (tested locally using Anvil and Foundry). After deployment, it functions as the “digital election officer”, overseeing the entire election process.

The blockchain workflow happens in four steps:

1. **Smart contract deployment:** The Election Commission deploys the *LocalVoting* contract, which sets the official rules of the election.
2. **Registration of candidates and voter commitments:** The Election Commission uses administrative functions in the contract to register public candidate identifiers and voter-eligibility commitments in the blockchain system.
3. **Vote casting:** Eligible voters cast their votes through the contract. The system ensures that each voter can vote only once.
4. **Result availability:** Anyone can later read the final vote counts directly from the blockchain, ensuring transparency and trust.

Because the blockchain is decentralized and tamper-proof, once recorded, vote-tally updates cannot be modified without altering the blockchain state, thereby improving integrity and auditability.

**Fig. 6** InsightFace face recognition process

2.4.1 Smart contract deployment

The smart contract is like a *rulebook written in computer code*. Once it is placed on the blockchain, no one can alter the rules, and every step follows the same procedure for everyone. In our system, we use only one contract named `LocalVoting`. It combines candidate registration, voter registration, voting, and result counting inside a single program.

- When the contract is deployed, the address of the deployer (the Election Commission) is stored as `electionCommission`.
- Only this address can register candidates, register voter commitments, and end the election. This is enforced by the `onlyCommission` modifier.

This deployment design makes the on-chain election workflow simple and robust. After deployment, the smart contract enforces registration, vote-state control, and tally updates deterministically. However, voter identity verification, salt storage, and the mapping between real identities and voter hashes remain handled by the protected off-chain infrastructure.

2.4.2 Pseudonymous vote submission & on-chain validation

In the proposed electronic voting framework, the voting phase is designed to enforce one-person-one-vote while reducing the direct disclosure of voter identity on the blockchain. Instead of submitting the voter's raw National ID (NID), the system derives a pseudonymous hash, $h_v = \text{Hash}(\text{NID}_v, s_v)$, where s_v is a unique random salt generated during the registration. During vote submission, the unique random salt s_v is retrieved by decrypting the stored encrypted salt with the secret key in the `.env` file. The voting process then proceeds as follows:

1. The voter is first authenticated through the IoT-assisted biometric pipeline (fingerprint or face recognition). Only after successful verification is the voter allowed to continue.
2. The backend retrieves the voter's off-chain identity record, recomputes the corresponding voter hash h_v , and prepares the ballot transaction using the selected public candidate identifier.
3. The voting application then calls the `castVote()` function of the smart contract with two inputs: the voter's pseudonymous hash and the chosen candidate identifier.
4. Before accepting the vote, the smart contract checks three conditions:

- whether the submitted voter hash corresponds to a registered eligible voter.
- whether the same voter hash has already been marked as used.
- whether the submitted candidate identifier is valid for the active election.

5. If all conditions are satisfied, the contract marks the voter hash as spent and increments the selected candidate's `voteCount` by one.

This mechanism improves privacy by keeping raw NIDs, encrypted salts, and biometric templates entirely off-chain and by using voter hash for on-chain eligibility verification and vote-status control. Within this design, each voter is represented on the blockchain by a derived pseudonymous identifier. At the same time, the mapping between the real identity and the corresponding hash is maintained only within the protected off-chain registration infrastructure. As a result, the public ledger does not directly disclose civil identifiers. The current system provides tamper-resistant tallying and one-person-one-vote enforcement, while stronger privacy guarantees, such as anonymous credentials, encrypted ballots, commitment-reveal schemes, or zero-knowledge proofs, remain part of future enhancements.

After the election ends, the number of votes received by each candidate can be verified directly from the smart contract state via each candidate's recorded `voteCount`. In addition, aggregate consistency can be cross-checked by comparing the sum of all candidates' `voteCount` values with the number of successful vote-casting records stored on-chain.

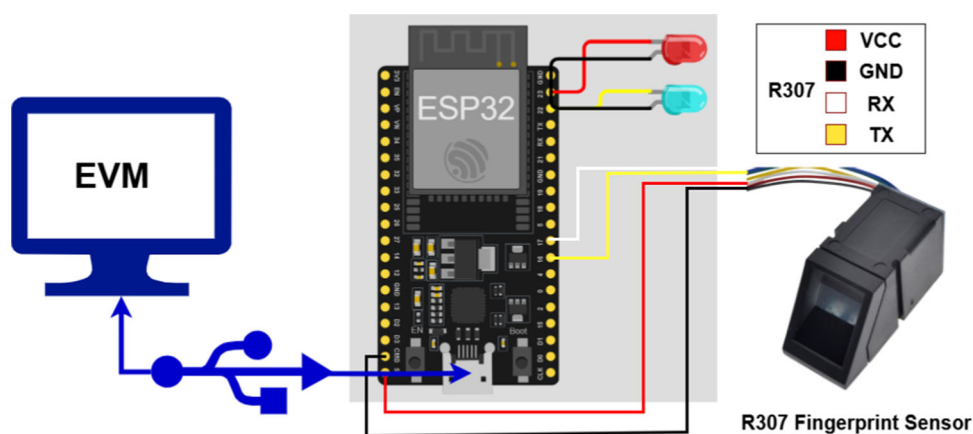
3 Results

This section presents the experimental setup and implementation outcomes of the electronic voting system prototype. We have discussed the hardware and software environment used in the experimental setup to evaluate the system on a small scale under realistic conditions. This setup provides the foundation for replicability, ensuring that results are produced in a controlled and well-defined environment. Finally, this section presents an analysis of the system's findings and performance.

3.1 Experimental setup

3.1.1 Hardware setup & software environment

The experimental evaluation of the proposed system was conducted in a controlled laboratory setting, utilizing a combination of IoT sensors and a blockchain-enabled server

Fig. 7 Circuit Diagram of IoT Hardware Setup

environment. The hardware and software components were carefully selected to simulate a realistic voting ecosystem, while ensuring the repeatability and robustness of the results. The hardware components used in this experiment are as follows:

- The ESP32 serves as a transceiver, sending biometric data to the server and relaying the response to the EVM.
- R307 optical fingerprint sensor is used to collect fingerprint images.
- Laptop's camera is used to capture live voters' facial images
- A laptop/Workstation is used as the main EVM central server, running the backend, blockchain node, and frontend application.

Figure 7 shows the circuit diagram of the hardware setup, including the ESP32 and the fingerprint sensor connected to the workstation. The connection details between the ESP32 and the R307 optical fingerprint sensor are shown in Table 3. The UART serial pins of the ESP32 were configured to communicate directly with the sensor's TX and RX pins, ensuring smooth template enrollment, feature extraction, and detection. Table 4 shows the workstation configuration and IoT hardware used in the experiment. On the other hand, the software stack comprises multiple programming languages, frameworks, biometric recognition modules, and blockchain tools. Table 5 lists the environment in which our proposed system was implemented and tested.

This hardware and software setup provided a controlled environment for conducting experiments with the proposed electronic voting framework, enabling end-to-end testing of

biometric verification, blockchain-based vote casting, and result tallying. This system includes a lightweight web-based application that connects voters, administrators, and the blockchain backend. The frontend, developed using React and TypeScript, provides two key interfaces: an admin dashboard for election management (including candidate registration, voter enrollment, and election activation) and a voter interface for biometric login and voting. Once a voter completes biometric verification through fingerprint or face recognition, the application automatically triggers a blockchain transaction via the MetaMask wallet to record the vote on the Ethereum network. A Public Dashboard summarizes live voting statistics and final tallies in real time, ensuring transparency without disclosing voter identities. This modular web interface integrates seamlessly with the IoT and blockchain layers through RESTful APIs, enabling end-to-end interaction across the system.

3.2 Result analysis

To evaluate our system's performance, we define a set of metrics covering communication latency, voting process efficiency, and biometric accuracy. The time required to transfer data from the IoT node (ESP32) to the server depends on the packet size and the communication link's baud rate. If S_{pkt} is the packet size in bits and R_{baud} is the baud rate in bits per

Table 3 Circuit connections between ESP32 and R307 Fingerprint Sensor

ESP32 Pin	5V/ Vin	GND	GPIO16 (RX2)	GPIO17 (TX2)
R307 Pin	VCC	GND	TX	RX

Table 4 Hardware setup used in the experimental evaluation

Category	Specification
Workstation (Server)	-Processor: Ryzen 7 5700U -Memory: 12 GB RAM -Storage: 512 GB SSD -Operating System: Windows 11 Home
IoT Hardware	ESP32 Microcontroller, R307 Fingerprint Sensor

Table 5 Software environment for proposed electronic voting system implementation

Layer	Technology
Frontend	React, TypeScript, Ethers.js
Wallet	Metamask
Backend	Node.js, Express.js
Face Recognition	Python, InsightFace, OpenCV, scikit-learn
Fingerprint Authentication	C#, .NET 8, SourceAFIS, CLI tools
Blockchain	Foundry, Anvil (Ethereum local chain)
Database	MongoDB

second, then the data transfer time is given by:

$$T_{\text{transfer}} = \frac{S_{\text{pkt}}}{R_{\text{baud}}}. \quad (4)$$

The average time required for a voter to complete the voting process is measured as the sum of the IoT data transmission time, biometric authentication time, liveness verification time, and blockchain confirmation time. If T_{transfer} denotes the biometric data transmission time from the IoT device to the server, T_{auth} denotes the biometric authentication time, T_{live} denotes the liveness detection time, and T_{block} denotes the blockchain confirmation time, then:

$$T_{\text{vote}} = T_{\text{transfer}} + T_{\text{auth}} + T_{\text{live}} + T_{\text{block}}. \quad (5)$$

The data transfer time between the IoT fingerprint module (ESP32 with R307 sensor) and the server was calculated using the data transfer time formula (4). In UART communication, each transmitted byte typically requires 10 bits, including 8 data bits, 1 start bit, and 1 stop bit. Therefore, for a fingerprint template packet size of $S_{\text{pkt}} = 36,864$ bytes and a baud rate of $R_{\text{baud}} = 57,600$ bps, the transfer time was computed as:

$$T_{\text{transfer}} = \frac{36864 \times 10}{57,600} = 6.4 \text{ seconds}.$$

By increasing the UART baud rate to 115,200 bps, this transmission time can be reduced to 3.2 seconds. This result indicates that fingerprint transmission accounts for a significant portion of the overall system delay. The voting time for each person was calculated using (5). Authentication using fingerprint or face recognition required approximately 6–8 seconds, while liveness detection (blinking and head-turn tests) required 4–6 seconds inside the authentication. The confirmation of blockchain transactions introduced an additional 1–2 seconds of delay on our local deployment.

Depending on network load and the time required for fingerprint data transfer, the end-to-end voting process per person takes an average of 10–15 seconds, with additional time overhead.

To evaluate the reliability of the biometric components, we analyzed the accuracy, False Acceptance Rate (FAR), and False Rejection Rate (FRR) of the face recognition module. The face recognition component was evaluated under controlled experimental conditions using the deep-funneled Labeled Faces in the Wild (LFW) dataset, which contains 13,233 images of 5,749 individuals, of whom 1,680 have two or more images. The evaluation was conducted in a pairwise face verification setting, where InsightFace embeddings were extracted from each image pair and cosine similarity was used to match them. The face recognition module achieved an overall accuracy of 96.09%, with FAR and FRR of 0% and 7.8%, respectively, using a threshold of 0.5. It is important to note that the reported FRR represents the false rejection rate per authentication attempt during dataset-based evaluation, where each comparison corresponds to a single verification instance, and should not be interpreted as the proportion of real voters rejected in a practical deployment. In the proposed system, multimodal biometric authentication is employed, allowing a voter to use fingerprint verification if face recognition fails alternatively, and to retry authentication if necessary. As a result, the effective rejection rate for legitimate voters in real-world use is expected to be lower than the reported FRR. The model demonstrated a best-case accuracy of 98.69% under optimized threshold (0.24) and feature extraction conditions. For fingerprint authentication, the SourceAFIS-based matcher was operated with a predefined FAR of 0.0001, from which the corresponding decision threshold was determined. Therefore, the reported FAR for the fingerprint module represents the selected operating point for threshold calibration, rather than an independently benchmarked fingerprint accuracy result.

People may try to perform duplicate voting in an electronic voting system. In the proposed framework, duplicate voting is prevented through a layered one-person-one-vote design that binds biometric authentication, off-chain identity handling, and on-chain eligibility control together. After successful biometric verification, the backend retrieves the voter's protected off-chain identity record, decrypts the stored salt, and recomputes the corresponding pseudonymous voter hash. Before accepting a new vote, the smart contract checks whether this derived voter hash corresponds to a registered eligible voter and whether the same voter hash has already been marked as used. If the voter hash has already been used, the transaction request is rejected immediately. In this way, the system prevents duplicate voting without exposing the voter's raw National ID on-chain, while preserving data integrity and trust in the electronic voting process. Additionally, every accepted transaction is immutably

recorded on the blockchain, which makes any tampering or bypass attempt immediately visible. This verification process ensures transparency and guarantees that the final tally reflects only legitimate and unique votes. Additionally, every voting transaction is immutably recorded on the blockchain, which makes any tampering or bypass attempt immediately visible. This strict verification process ensures transparency and guarantees that the final tally reflects only legitimate, unique votes.

3.3 Blockchain performance analysis

The blockchain layer in our system ensures transparency, immutability, and verifiability throughout the electronic voting process. To evaluate its computational efficiency, a detailed performance analysis was conducted using the Foundry framework on a local Anvil blockchain network. This section presents the variations in gas consumption, deployment cost, and transaction cost for the `LocalVoting` smart contract under different network conditions.

3.3.1 Smart contract deployment results

The smart contract `LocalVoting.sol` was compiled, deployed, and tested in the Foundry environment. Figure 8 shows the summary output of the `forge test --gas-report` command. The deployment consumed approximately **1,891,738 gas units**, producing a contract binary size

of **8,464 bytes**. The functions `registerCandidate()` and `registerVoter()` exhibited the highest gas usage because they update multiple mappings and arrays on the blockchain. In contrast, view-type functions such as `getCandidate()` and `getVoter()` consumed fewer than 6,500 gas units, confirming their efficiency.

3.3.2 Ethereum gas price variability

To simulate realistic blockchain conditions, daily average gas-price data were collected from Etherscan (via YCharts) over the period **September 3 to October 22, 2025**. As shown in Fig. 9, the Ethereum network experienced varying gas prices, ranging between **0.86 Gwei** and **18.2 Gwei**, with percentile values of **p10 = 0.843 Gwei**, **p50 = 1.427 Gwei**, and **p95 = 3.585 Gwei**. These values were adopted as the *Low*, *Typical*, and *Stress* scenarios for cost evaluation.

3.3.3 Gas-cost sensitivity analysis

Using the gas-usage metrics from Foundry and the recorded gas-price scenarios, the transaction cost (in USD) of each smart-contract function was calculated according to:

$$\text{Cost (USD)} = \text{GasUsed} \times \text{GasPrice(Gwei)} \times 10^{-9} \times \text{ETHPrice(USD)}.$$

The exchange rate was fixed at **\$3,890 per ETH** for reproducibility and comparability with other works. Figure 10

src/LocalVoting.sol:LocalVoting Contract					
Deployment Cost	Deployment Size				
1891738	8464				
Function Name	Min	Avg	Median	Max	# Calls
castVote	25551	52221	53461	76441	6
electionEnded	393	1393	1393	2393	2
endElection	23567	25009	25009	26451	2
getCandidate	6228	6228	6228	6228	2
getVoter	5845	5845	5845	5845	1
registerCandidate	29648	187704	214047	214047	7
registerVoter	29472	166367	189173	189233	7

Fig. 8 Foundry-based gas consumption report of the `LocalVoting` smart contract

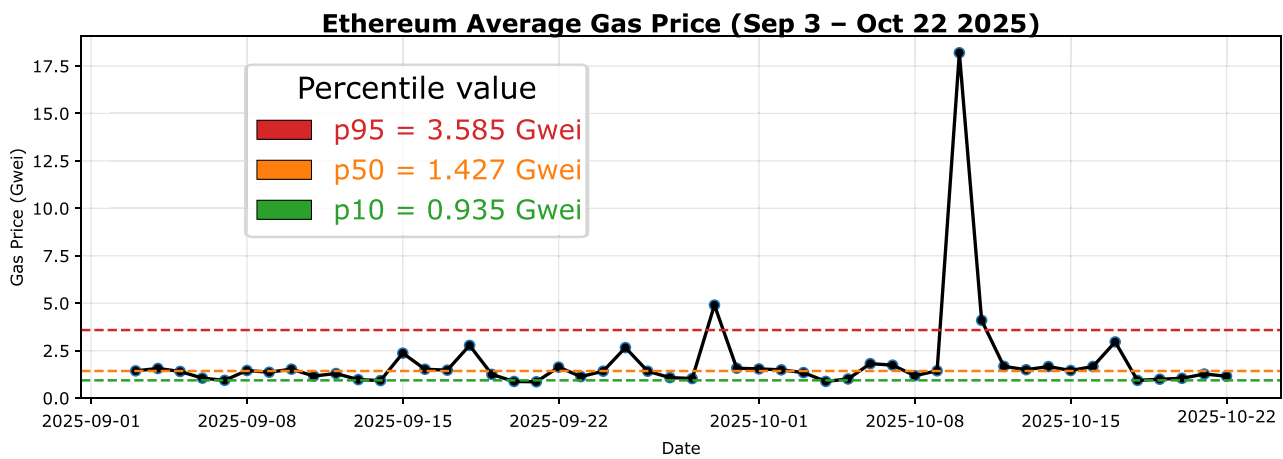


Fig. 9 Historical average gas price (Gwei) from September 3 to October 22, 2025, sourced from ycharts.com

presents the resulting transaction costs across four gas conditions: Low (1 Gwei), Typical (1.45 Gwei), Stress (3.585 Gwei), and Baseline (20 Gwei). The functions `registerCandidate()` and `registerVoter()` dominate the cost distribution, whereas read-only functions remain below \$0.50 even under stress conditions.

3.3.4 Quantitative comparison

Table 6 summarizes the transaction costs of each function under the four gas-price scenarios. The results show that, under typical network conditions (around 1.45 Gwei), executing all contract functions costs less than 1 U.S. dollar.

3.3.5 Deployment-cost evaluation

The one-time deployment of the `LocalVoting` contract required approximately 1.89 million gas units, corresponding to a deployment cost of **\$113.50** under the 20 Gwei baseline. Table 7 summarizes the deployment cost, total gas consumption, and contract size of the proposed smart contract under these baseline conditions. This cost is relatively small for public election authorities, confirming the cost-efficiency of our system's on-chain architecture. The blockchain performance evaluation confirms that the proposed system achieves low operational overhead while maintaining the immutability and transparency expected of a decentralized system. Under typical network conditions

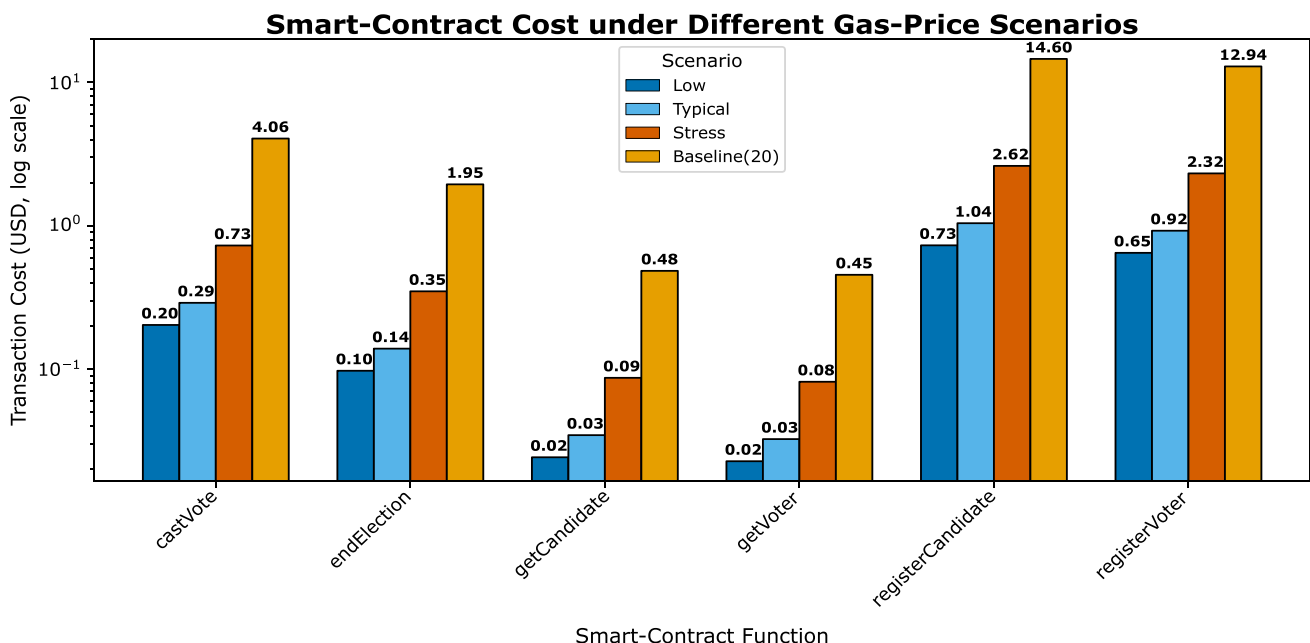


Fig. 10 Transaction cost of the proposed electronic voting system's smart-contract functions under different gas price scenarios

Table 6 Average execution cost of the proposed system's methods (in USD)

Method	Baseline (20 Gwei)	Low (1 Gwei)	Stress (3.585 Gwei)	Typical (1.45 Gwei)
castVote	4.0628	0.2031	0.9954	0.2946
endElection	1.9457	0.0973	0.4767	0.1411
getCandidate	0.4845	0.0242	0.1187	0.0351
getVoter	0.4547	0.0227	0.1114	0.0330
registerCandidate	14.6034	0.7302	3.5778	1.0587
registerVoter	12.9434	0.6472	3.1711	0.9384

(1.45 Gwei), all core contract functions cost less than 1 U.S. dollar, while the deployment cost remains within affordable limits for government-scale use. The results validate the system's suitability as a practical, cost-efficient, and scalable e-voting platform.

4 Threat model and security goals

To clarify the security scope of the proposed electronic voting framework, this section defines the system boundary, trust assumptions, adversarial capabilities, security goals, and limitations. The threat model focuses on attacks involving voter authentication, biometric verification, off-chain identity management, pseudonymous blockchain interactions, and smart-contract-based vote recording.

4.1 System boundary

The proposed system consists of four main components: the IoT biometric acquisition layer, the off-chain biometric and identity management layer, the application/backend layer, and the blockchain-based vote recording layer. The IoT layer collects fingerprint and facial data during voter registration and vote casting. The off-chain layer stores sensitive records, including raw NID values, biometric data, fingerprint templates, face embeddings, and encrypted salts. The application/backend layer coordinates biometric verification, voter hash reconstruction, and vote submission requests. The blockchain layer stores registered voter hashes, candidate identifiers, vote status information, and candidate vote counts via the smart contract.

This boundary separates sensitive identity and biometric processing from public blockchain storage. Raw voter

identifiers and biometric records remain inside the protected off-chain infrastructure, while the blockchain receives only the derived pseudonymous voter hash for eligibility verification and duplicate-vote prevention.

4.2 Trust assumptions

The proposed system is based on the following trust assumptions:

- The blockchain layer is assumed to provide tamper-resistant storage for accepted vote transactions and candidate tally updates after the vote has been recorded.
- The deployed smart contract is assumed to execute according to its implemented logic. In the proposed prototype, the smart contract controls candidate registration, voter-hash registration, vote casting, duplicate-vote prevention, and result counting.
- The Election Commission or authorized administrative authority is assumed to be responsible for voter registration, candidate registration, election activation, and election termination.
- The admin wallet and smart-contract access-control logic are assumed to restrict administrative operations to authorized election officials.
- The off-chain identity database, backend server, and biometric verification servers are considered partially trusted components. They store or process sensitive data and may therefore become attack surfaces if misconfigured, compromised, or misused by insiders.
- The secret key used for salt encryption and decryption is assumed to remain protected within the backend environment. Exposure of this key would weaken the protection of the stored encrypted salts.
- Communication between IoT devices, backend services, biometric verification modules, and blockchain interfaces must be protected in practical deployment. The current prototype focuses on system implementation and functional evaluation, while deployment-grade communication hardening remains necessary for real-world use.

Table 7 Deployment cost of the LocalVoting smart contract under baseline conditions (20 Gwei)

Smart Contract	Deployment Cost (USD)	Deployment Gas	Contract Size (bytes)
LocalVoting	113.50	1,891,738	8,464

- **IoT biometric devices** are assumed to capture voter biometric inputs during registration and authentication. The current prototype does not claim complete resistance against advanced physical tampering, sensor replacement, or hardware-level side-channel attacks.

4.3 Adversary model

We consider the following adversarial entities:

- **External adversary:** An unauthorized attacker who may attempt to intercept communication, submit invalid vote requests, impersonate voters, or disrupt system availability.
- **Malicious voter:** A registered voter who may attempt to vote more than once, use another voter's identity, bypass biometric verification, or manipulate the client-side voting interface.
- **Compromised biometric component:** A faulty or compromised biometric verification server that may return an incorrect identity, reject a legitimate voter, or attempt to influence the authentication result.
- **Insider adversary:** A privileged operator, administrator, or election official who may attempt to manipulate voter records, biometric records, encrypted salts, candidate records, or verification outputs before they are submitted to the blockchain layer.
- **Network adversary:** An attacker who may attempt to observe, delay, replay, or modify communication between IoT devices, backend services, biometric modules, and blockchain interfaces.
- **Physical or side-channel adversary:** An attacker who may target IoT biometric devices or processing units through physical tampering, timing leakage, communication-pattern analysis, or sensor-level manipulation.

4.4 Security goals

Under the above assumptions and adversary model, the proposed system aims to achieve the following security goals:

- **Authentication integrity:** Only registered voters who successfully pass biometric verification should be allowed to proceed to the vote-casting stage.
- **Robust biometric verification:** The system should reduce dependence on a single biometric verification point by accepting a voter identity only when the required majority-consensus threshold is satisfied across multiple biometric verification servers.
- **Resistance to double voting:** Each derived pseudonymous voter hash should be allowed to cast at most one accepted vote. Once the smart contract marks a voter hash

as used, any further vote attempts using the same hash must be rejected.

- **Pseudonymous identity protection:** Raw NID values, biometric records, fingerprint templates, face embeddings, and encrypted salts should not be directly stored on-chain. The blockchain should use only the derived pseudonymous voter hash for eligibility verification and vote-status control.
- **Vote-record integrity:** Once the smart contract accepts a vote, the corresponding vote-status update and candidate tally should remain tamper-resistant through the blockchain state.
- **Public tally transparency:** Aggregate vote counts should be visible through the public result dashboard or directly from the blockchain state without requiring access to raw voter identities or biometric records.
- **Limited public unlinkability:** Public blockchain observers should not be able to directly infer a voter's raw civil identity from the on-chain voter hash alone. However, this property is limited because the protected off-chain authority maintains the voter identity record and encrypted salt needed to associate a voter with the corresponding pseudonymous hash.

4.5 Security considerations and limitations

The blockchain layer provides tamper-resistant vote recording after a vote has been accepted. However, blockchain alone does not protect against all attacks originating from off-chain components. In particular, biometric servers, IoT devices, administrative interfaces, backend services, and off-chain databases remain important attack surfaces. Insider adversaries may attempt to manipulate voter records or biometric verification outputs before submitting to the blockchain. The proposed system reduces this risk through admin-wallet-based authorization, smart contract access control, off-chain protection of sensitive records, pseudonymous voter hashes for blockchain interactions, and majority-consensus biometric verification across multiple servers. However, these mechanisms do not eliminate all risks associated with off-chain compromise or insider misuse. This pseudonymity has several limitations and should not be interpreted as full voter anonymity. In the current architecture, the public blockchain stores and processes only the derived voter hash. At the same time, raw NID values, biometric records, and encrypted salts remain within the protected off-chain infrastructure. This design prevents direct exposure of identities on the public ledger. Nevertheless, because the off-chain authority retains the voter identity record and the encrypted salt, it can maintain or reconstruct the association between a real voter and the corresponding pseudonymous hash when required. Therefore, although the proposed approach reduces public identity

disclosure, it does not provide full cryptographic voter anonymity, coercion resistance, receipt-freeness, or complete ballot secrecy. Achieving these stronger privacy properties would require additional cryptographic mechanisms, such as anonymous credentials, encrypted ballots, mixnets, commit-reveal protocols, or zero-knowledge proofs, which are identified as future extensions of the proposed framework. In addition, the current prototype does not fully address advanced physical attacks, sensor replacement attacks, large-scale denial-of-service attacks, endpoint compromise, or hardware-level side-channel leakage. Practical deployment would require stronger communication security, device hardening, secure key management, administrative auditing, and larger-scale security testing.

5 Discussion

Our proposed electronic voting system is a successful integration of low-cost IoT devices and multi-modal biometric authentication with the blockchain-based e-voting system. During the design, development, and testing phases, we encountered challenges at the hardware, communication, and software levels. We used a low-cost R307 fingerprint sensor with an ESP32 microcontroller. The default Adafruit library stores fingerprint templates internally, limiting user control over feature extraction and matching logic. Once the fingerprint sensor becomes deprecated, the whole fingerprint database will be inaccessible. To mitigate this issue, we have reset its firmware using microPython and modified the pyFingerprint library to capture raw fingerprint images directly and store them outside the sensor's memory. This intervention helped to achieve both flexibility and data portability, ensuring necessary hardware-level improvements.

The communication latency between the IoT fingerprint devices (ESP32 with R307 sensor) and the server is significantly affected by the UART baud rate and transmission overhead. Considering UART framing, where each byte typically requires 10 bits for transmission, a fingerprint template packet size of $S_{\text{pkt}} = 36,864$ bytes corresponds to an effective transmission size of 368,640 bits. At the initial baud rate of $R_{\text{baud}} = 57,600$, the transmission time is

approximately 6.4 seconds. By increasing the baud rate to 115,200, this delay can be reduced to approximately 3.2 seconds. This shows that fingerprint data transmission accounts for a significant portion of the total voting time, and that improving communication settings can substantially enhance overall system responsiveness. Consequently, the average end-to-end voting time per person was approximately 10–15 seconds, which is acceptable for small- to medium-scale elections. However, further optimization would be needed for large-scale deployments.

Fingerprint authentication was performed using a Source-AFIS-based matcher configured with a target FAR of 0.0001, from which the corresponding matching threshold was derived. To ensure the accuracy of the face recognition model evaluation, we conducted a controlled benchmark using the Labeled Faces in the Wild (LFW) dataset. With a similarity threshold of 0.5, the face recognition module achieved an overall accuracy of 96.09%, a FAR of 0%, and a FRR of 7.8%. Under optimized threshold (0.2431) and feature-extraction conditions, the model also demonstrated a best-case accuracy of 98.69%. These results indicate that the face recognition component is reliable under controlled conditions. However, in real-world environments, its performance may vary. We couldn't collect sufficient facial data in a real-world environment. This represents another limitation of the current work, and future experiments will need to gather a larger, more diverse biometric dataset of real voters to evaluate true recognition performance under realistic conditions.

Beyond biometric performance, the proposed system also enhances privacy and anti-spoofing at the system design level. Sensitive records such as raw NID numbers, biometric templates, and encrypted salts are stored within the protected off-chain infrastructure. At the same time, blockchain interactions use a derived pseudonymous voter hash rather than the voter's NID number. This design reduces direct exposure of identities on the public ledger and improves privacy compared with architectures that submit voter identifiers directly on-chain. In addition, integrating liveness-aware facial verification via blinking and head-movement checks helps mitigate presentation attacks and strengthens resistance to biometric spoofing during authentication.

However, this pseudonymity has several limitations and should not be interpreted as full voter anonymity. In the cur-

Table 8 Cost analysis of hardware setup for the proposed system

Category	Component	Quantity	Unit Cost (USD)	Total Cost (USD)
Workstation (Server)	Laptop (Ryzen 7 5700U, 12GB RAM, 512GB SSD)	2	660	1320
IoT Hardware	ESP32 microcontroller	1	3	3
Sensor	R307 fingerprint sensor	1	15	15
Connection	USB Serial Cable	1	2	2
Total Project Cost				1340

Table 9 Comparative analysis of existing blockchain-based e-voting system with our proposed system

System	Blockchain Type	Smart Contracts	Biometrics	IoT	Anti spoofing Methods	Privacy-preserving Methods	Gas Cost
Indrason et al. (2025)	Consortium blockchain	No	Yes	Yes	Not mentioned	Yes (basic)	Cost not measured
Razaque et al. (2025)	Consortium blockchain	Yes	Yes (partially)	No	Not mentioned	Yes	Low gas cost
Somasekhar et al. (2024)	Public blockchain (Ethereum)	Yes	Yes (partially)	No	Not mentioned	Yes (basic)	High gas costs
Jayakumari et al. (2024)	Hybrid blockchain	Yes	Yes (partially)	Yes	Not mentioned	Yes (basic)	Cost not analyzed
Hossain Faruk et al. (2024)	Private blockchain (Hyperledger Fabric)	Yes	Yes	No	Not mentioned	Yes (basic)	Cost not analyzed
Our proposed framework	Public blockchain (Ethereum)	Yes	Yes–Multi modal (Fingerprint, face)	Yes	Yes	Yes	Medium gas cost

rent architecture, the public blockchain stores and processes only the derived voter hash. At the same time, raw NID values, biometric records, and encrypted salts remain within the protected off-chain infrastructure. This design prevents direct exposure of identities on the public ledger. Nevertheless, because the off-chain authority retains the voter identity record and the encrypted salt, it can maintain or reconstruct the association between a real voter and the corresponding pseudonymous hash when required. Therefore, although the proposed approach reduces public disclosure of voter identities, it does not provide full cryptographic voter anonymity or complete ballot secrecy. Achieving these stronger privacy properties would require additional cryptographic mechanisms, such as anonymous credentials, encrypted ballots, commit-reveal protocols, and zero-knowledge proofs, which are identified as future extensions of the proposed framework.

The system's scalability and blockchain optimization remain important directions for future research. Large-scale stress testing with thousands of concurrent voters is still needed to evaluate readiness for real-world election deployment. Exploring alternative blockchain frameworks, such as Hyperledger Fabric or Tendermint, or adopting Layer-2 scaling techniques could further reduce operational cost and improve throughput. In addition, stronger privacy-preserving cryptographic mechanisms, such as zero-knowledge proofs (zk-SNARKs or zk-STARKs), anonymous credentials, or encrypted-ballot techniques, may further enhance ballot secrecy and unlinkability in future versions of the system. Despite these limitations, the proposed framework demonstrates how multimodal biometric verification, privacy-preserving pseudonymous identity handling, and blockchain-based vote validation can jointly strengthen the integrity of e-voting. Sensitive records, including raw NID values, biometric templates, and encrypted salts, remain protected within the off-chain infrastructure. At the same time, blockchain interactions rely on a derived pseudonymous voter hash rather than direct exposure of civil identity. Furthermore, integrating fingerprint and face recognition with liveness-aware anti-spoofing checks reduces the risk of impersonation and presentation attacks. The system also addresses duplicate voting through a one-person-one-vote design, enforced by both biometric re-verification and on-chain smart contract checks. As a result, once a voter's hash is marked as used, any subsequent attempt associated with the same verified identity is automatically rejected.

Additionally, we analyzed the hardware costs of implementing this system at a small scale. Table 8 presents the cost distribution of the main components used. Although the workstation represents a significant expense, the IoT devices are very cost-effective, and the system is feasible for deployment in real-world scenarios. For better clarity, we have also compared our proposed system with several existing works

using the criteria listed in Table 9. The results show that our system fulfills all the requirements considered in the comparison.

6 Conclusion

The development of the proposed electronic voting system demonstrates how integrating biometric authentication and blockchain technology can address long-standing challenges in electronic voting. By combining fingerprint verification with InsightFace-based facial recognition enhanced by liveness detection, the system provides strong identity validation and significantly reduces the risk of impersonation or spoofing attacks. On the security side, the blockchain layer ensures immutability, transparency, and verifiability of votes, thereby building trust among stakeholders by preventing unauthorized manipulation of election data. From an implementation perspective, the project successfully bridges hardware and software through an ESP32 microcontroller connected to the R307 fingerprint sensor, an Express.js backend for reliable data handling, a React/TypeScript frontend for user interaction, and Foundry-based smart contracts for secure, decentralized vote recording. This end-to-end prototype demonstrates how multiple emerging technologies can be orchestrated into a coherent system that not only ensures integrity and accuracy but also remains scalable and user-friendly. While the current prototype provides a strong foundation, it is essential to acknowledge its limitations. Features such as large-scale performance testing, advanced cryptographic enhancements, and compliance with legal and regulatory frameworks are still pending and represent the next stage of development. Nonetheless, the successful realization of our system at this stage proves the feasibility of a biometrically authenticated, blockchain-enabled e-voting system that enhances trust, security, and efficiency in democratic processes. Ultimately, this work contributes to the vision of next-generation digital voting platforms that are tamper-resistant, inclusive of voters, and capable of safeguarding transparency and privacy in modern democracies.

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Author Contributions Tahzib Mahmud Rifat and Md Numanur Rahman contributed to the conceptualization of the study, methodology design, data collection, and initial experimentation. Saurav Chandra Das was responsible for supervision, research design refinement, formal analysis, and overall project administration. Mahir Mahbub contributed to software development, model implementation, and result validation. Md. Sanaullah and Md. Ashiquesalehin assisted in data preprocessing, experimentation, and performance evaluation. Saad Aloteibi contributed to supervision, critical review, and editing of the manuscript. Mohammad Ali Moni provided high-level supervision, methodological guidance, and manuscript review and editing.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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Authors and Affiliations

Tahzib Mahmud Rifat¹ · Md. Numanur Rahman¹ · Saurav Chandra Das¹ · Mahir Mahbub¹ · Md. Sanaullah² · Md. Ashiqussalehin¹ · Saad Aloteibi³ · Mohammad Ali Moni^{4,5}

✉ Mohammad Ali Moni
mmoni@csu.edu.au

Tahzib Mahmud Rifat
2001005@iot.uftb.ac.bd

Md. Numanur Rahman
2001011@iot.uftb.ac.bd

Saurav Chandra Das
saurav0001@uftb.ac.bd

Mahir Mahbub
mahbub0001@uftb.ac.bd

Md. Sanaullah
sanaullah@ihc.jnu.ac.bd

Md. Ashiqussalehin
ashiqussalehin0001@uftb.ac.bd

Saad Aloteibi
SaadAloteibi@ksu.edu.sa

¹ Department of Internet of Things and Robotics Engineering, University of Frontier Technology, Bangladesh, Kaliakair, Gazipur, Bangladesh

² Jagannath University, Dhaka, Bangladesh

³ Department of Computer Science and Engineering, College of Applied Studies, King Saud University, 11437 Riyadh, Saudi Arabia

⁴ AI & Digital Health Technology, Artificial Intelligence and Cyber Futures Centre, Charles Sturt University, Bathurst, NSW 2795, Australia

⁵ AI & Digital Health Technology, Rural Health Research Institute, Charles Sturt University, Orange, NSW 2800, Australia